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0.21 | Calculate Bearing Shape Factor

The stiffness of rubber in compression, when the loaded surfaces are prevented from slipping (by bonding or by mechanical location), depends upon the shape factor S (see Fig. 1), defined as the ratio of one loaded area to the total force-free area. [0.2.1]

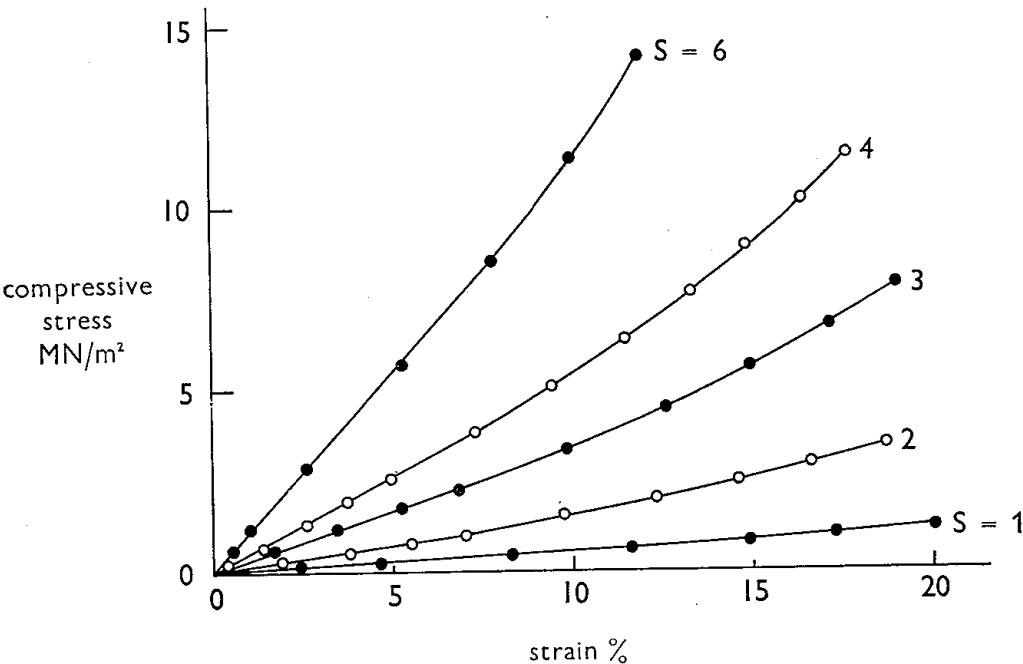


Fig. 27.—Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.

Fig. 1 - Shape Factor

Effect of shape factor: Experimental stress-strain curves for 6.3 mm thick disks of rubber (47 IRHD) in compression. The shape factor is shown alongside each curve; the diameter in mm is 25.4 times the shape factor.



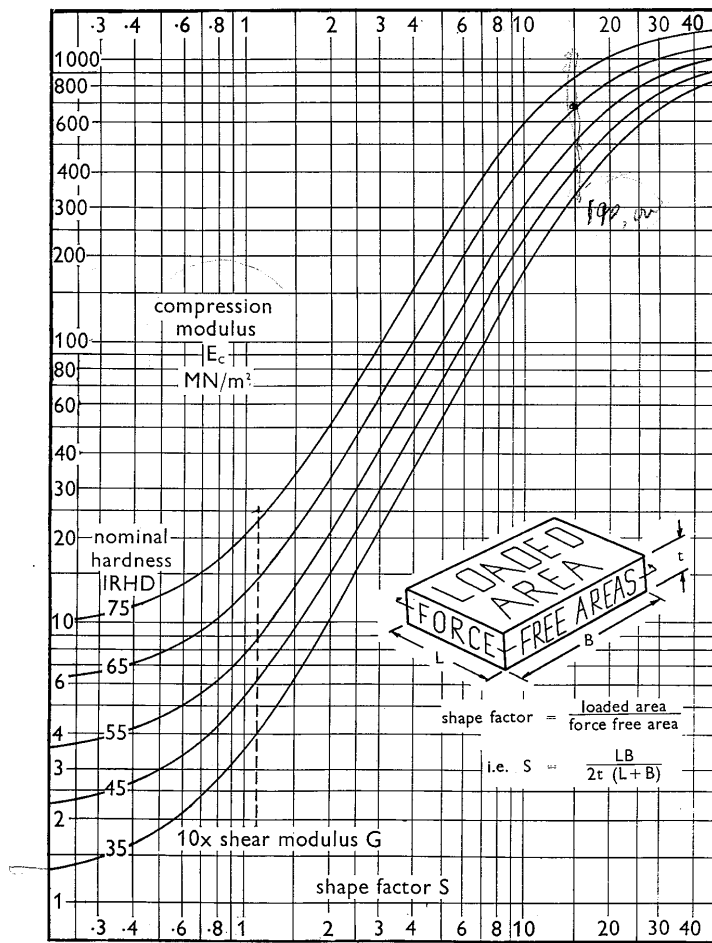


Fig. 28.—Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above).

* Although deformation due to bulk compression can normally be ignored, it can cause a noticeable reduction in E_c when the ratio E_c/E_∞ exceeds about 10%. (E_∞ is the modulus of bulk compression, from Table 3.) To allow for this reduction, use a modified compression modulus obtained by dividing E_c by $1+(E_c/E_\infty)$. If Wood and Martin's value for bulk modulus is preferred (see footnote to Table 2) then the E_∞ value in Table 3 should be doubled.

Fig. 2 - Compression Modulus

Variation of compression modulus E_c with shape factor S for natural rubbers of differing hardnesses (SRF black filler used for 55 IRHD and above). Chart shows compression modulus E_c (MN/m²) on the vertical axis (log scale, 1 to 1000) versus shape factor S on the horizontal axis (log scale, 0.3 to 40), with separate curves for nominal hardness values of 35, 45, 55, 65, and 75 IRHD, and a line labeled "10x shear modulus G ". An inset diagram illustrates a rubber block showing the "loaded area" (top and bottom faces) versus "force free areas" (side faces), with shape factor = loaded area / force free area, i.e. $S = (LB) / (2t*(L+B))$.

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0.2 - 22 | Modulus

$$S = (LB) / (2t*(L+B))$$

S = shape factor t = thickness L = length B = breadth

For a block of square section (i.e. $L = B$) or circular section (diameter = L):

$$S = L / (4*t)$$

The compression modulus E_c depends upon the shape factor S (for derivation see Appendix).

$$E_c = E_0(1 + 2kS^2)$$

E_c = compression modulus E_0 = Young's modulus (from Table 3) S = shape factor k = a numerical factor (from Table 3)

See footnote on following page.

When $S > 3$ it may be more convenient to use

$$E_c \approx 5GS^2$$

where G is the shear modulus (from Table 3).

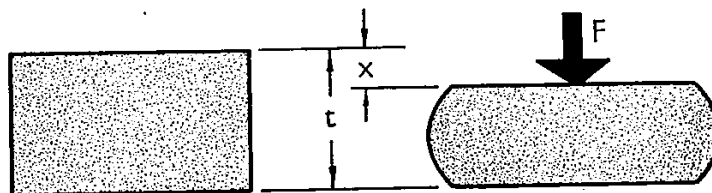


Fig. 29.—Rubber block in compression.

Fig. 3 - Compression Stiffness K_c

Rubber block in compression. Diagram shows a rectangular rubber block of thickness t , deflecting by x under a compressive load F applied to a cylindrical rubber sample.

The compression stiffness, K_c , is given by

$$K_c = F/x = E_c A/t$$

where F = compressive load E_c = compression modulus (corrected, if necessary, for the effect of bulk compression) A = cross-sectional area t = thickness x = deflexion

It is not advisable to rely on friction alone to prevent slip under a compressive load when using unbonded rubber parts, because slip may occur if $S > \mu/2$, where S is the shape factor and μ the coefficient of friction.



The load (F)—deflexion (x) curve of rubber in compression is non-linear. With no slip it has the approximate form

$$F = E_c A_e (1 + e)$$

where e, the compressive strain, equals x/t. The non-linearity is usually ignored for strains up to about 10%.

There is as yet no method of calculating that a block will be stable but experience has shown that provided the thickness is less than one-quarter of the least plan dimension there should be no instability.

0.2 - 33 | Strip Analysis

COMPRESSION STRIP

When a long strip of rubber is compressed the strain in the direction of its length will be negligible.

Shape factor

$$S = b / (2 \cdot t)$$

IMAGE | [img/fig30.png](#) | Compression Strip, 80, num, not

Diagram shows a long rectangular strip of rubber of width b, thickness t, and unit length, compressed by a load F applied per unit length.

Compression modulus E_c

$$E_c = 4 \cdot E_0 (1 + k \cdot S^2) / 3$$

The compression stiffness per unit length, K_c , is given by

$$K_c = F/x = E_c \cdot b/t = (4 \cdot b \cdot E_0 (1 + k \cdot S^2)) / (3 \cdot t)$$

where F = load per unit length E_c = compression modulus (corrected, if necessary, for the effect of bulk compression) E_0 = Young's modulus (from Table 3) b = width of strip t = thickness of strip x = deflexion k = a numerical factor (from Table 3) S = shape factor

The load per unit length (F)—deflexion (x) curve for a compressed strip is non-linear. It has the approximate form

$$F = E_c \cdot b \cdot e (1 + 3 \cdot e/2)$$

where e, the compressive strain, equals x/t. As in the case of blocks, the non-linearity is usually ignored for strains up to about 10%.

[0.2.1]

P.B. Lindley, Engineering Design with Natural Rubber, NR Technical Bulletin. Malaysian Rubber Producers' Research Association, Brickendonbury, U.K.

